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Faculty of Mathematics and Computer Science
Institute of Computer Science
Research Group: *<to be defined>*

Bachelor's Thesis in Computer Science

Data-driven camera calibration for light field camera systems

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Abstract

[1]

Contents

1. Introduction	1
1.1. Motivation	1
1.2. Challenges and Problem Statement	1
1.3. Research Questions	1
1.4. Structure of the Thesis	1
2. Theoretical Background	2
2.1. Pinhole Camera Model	2
2.2. Lens Distortion	2
2.2.1. Radial Distortion	2
2.2.2. Tangential Distortion	2
2.3. Camera Calibration	2
2.3.1. Calibration Parameters and Error Metrics	2
2.3.2. Target-Based Camera Calibration	2
2.3.3. Bundle Adjustment	2
2.4. Light Field Camera Systems	2
2.4.1. Multi-Camera Light Field Systems	2
2.4.2. Micro-Lens-Array-Based Light Field Cameras	2
2.5. State of the Art in Light Field Camera Calibration	2
3. Material and Method	3
3.1. Material	3
3.1.1. Synthetic Data Generation with Blender	3
3.1.2. Image Sets and Ground Truth Definition	3
3.1.3. Runtime Environment and Toolchain	3
3.2. Method	3
3.2.1. Use Case Description	3
3.2.2. Requirements Analysis	3
3.2.3. UML Class Diagram	3
3.2.4. Health Monitoring and Decision Metrics	3
3.2.5. State Diagram	3
4. Results and Evaluation	4
4.1. Results of Initial Calibration	4
4.1.1. Intrinsic Accuracy	4
4.1.2. Extrinsic Accuracy	4
4.1.3. Reprojection and RMS Error	4
4.2. Results of LiFCal-Based Recalibration	4
4.2.1. Reprojection Error after LiFCal	4
4.2.2. Health Score Evolution	4
5. Discussion of the Results	5
5.1. Interpretation of Initial Calibration Results	5
5.2. Limitations of LiFCal in the Given Setup	5
5.3. Comparison between Initial Calibration and LiFCal	5
5.4. Comparison to state of the Art	5
5.5. Potential Extensions and Improvements	5

6. Conclusion and Outlook	6
6.1. Conclusion	6
6.2. Outlook	6
A. Appendix	6

List of Figures

1. Introduction

1.1. Motivation

1.2. Challenges and Problem Statement

1.3. Research Questions

The following research questions guide this thesis:

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

1.4. Structure of the Thesis

2. Theoretical Background

2.1. Pinhole Camera Model

2.2. Lens Distortion

2.2.1. Radial Distortion

2.2.2. Tangential Distortion

2.3. Camera Calibration

2.3.1. Calibration Parameters and Error Metrics

2.3.2. Target-Based Camera Calibration

2.3.3. Bundle Adjustment

2.4. Light Field Camera Systems

2.4.1. Multi-Camera Light Field Systems

2.4.2. Micro-Lens-Array-Based Light Field Cameras

2.5. State of the Art in Light Field Camera Calibration

3. Material and Method

3.1. Material

3.1.1. Synthetic Data Generation with Blender

3.1.2. Image Sets and Ground Truth Definition

3.1.3. Runtime Environment and Toolchain

3.2. Method

3.2.1. Use Case Description

3.2.2. Requirements Analysis

3.2.3. UML Class Diagram

3.2.4. Health Monitoring and Decision Metrics

3.2.5. State Diagram

4. Results and Evaluation

4.1. Results of Initial Calibration

4.1.1. Intrinsic Accuracy

4.1.2. Extrinsic Accuracy

4.1.3. Reprojection and RMS Error

4.2. Results of LiFCal-Based Recalibration

4.2.1. Reprojection Error after LiFCal

4.2.2. Health Score Evolution

5. Discussion of the Results

5.1. Interpretation of Initial Calibration Results

5.2. Limitations of LiFCal in the Given Setup

5.3. Comparison between Initial Calibration and LiFCal

5.4. Comparison to state of the Art

5.5. Potential Extensions and Improvements

6. Conclusion and Outlook

6.1. Conclusion

6.2. Outlook

A. Appendix

Acknowledgment

References

- [1] Aymeric Fleith, Doaa Ahmed, Daniel Cremers, and Niclas Zeller. Lifcal: Online light field camera calibration via bundle adjustment. <https://arxiv.org/abs/2408.11682>, 2024.